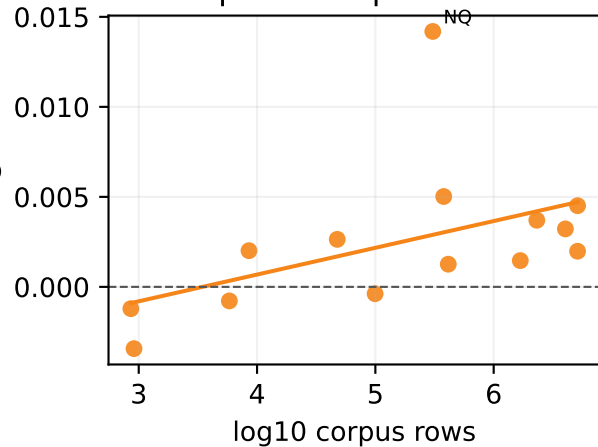
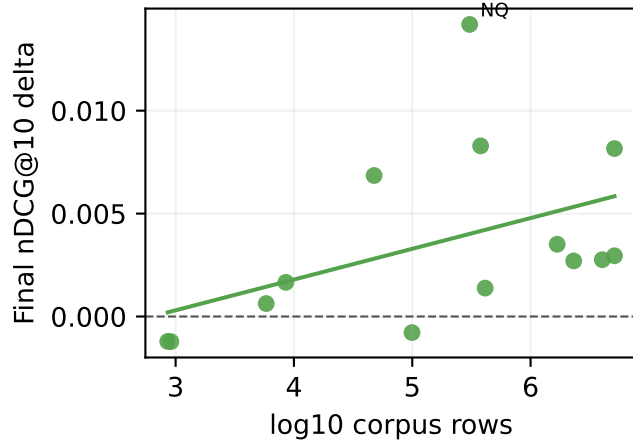


Exploratory optimizer gains versus full-corpus size

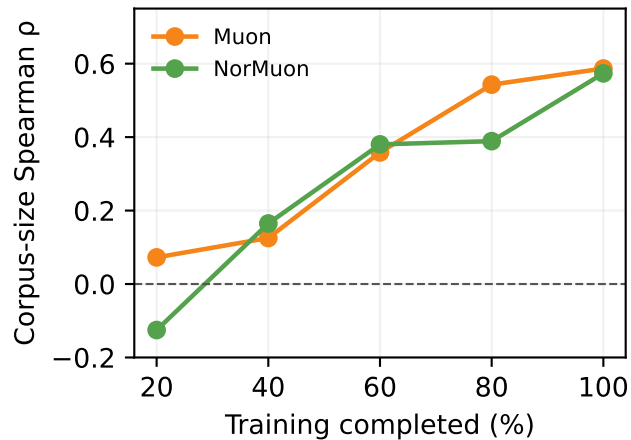
Muon – AdamW
Spearman $\rho=0.587$



NorMuon – AdamW
Spearman $\rho=0.574$



Association emerges late



Post-hoc, same-suite-selected discovery runs; association is descriptive, not causal.